3.3 Rotations

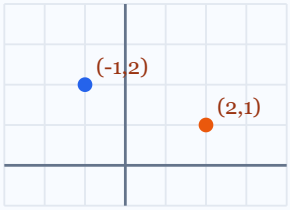
A rotation turns every point of a figure through the same angle about a fixed center, which is why every image point stays exactly as far from the center as its preimage did. Beyond the three memorized rules for rotating about the origin (90°, 180°, 270°), this section extends rotation to a center that is *not* the origin (translate the center to the origin, rotate, translate back), works backward to identify an unknown rotation angle from a point and its image, and previews how repeated rotations connect to rotational symmetry in Section 3.5.

Examples

Example 1: 90° counterclockwise about the origin

Rotate $(2, 1)$ 90° CCW about the origin, and explain where the rule $(x, y) \to (-y, x)$ comes from.

Solution



A quarter-turn CCW about the origin swaps coordinates and negates the new x.

Rule: a 90° CCW rotation about the origin sends $(x, y) \to (-y, x)$ — swap the coordinates, then negate the new x.

Apply it: $(2, 1) \to (-1, 2)$.

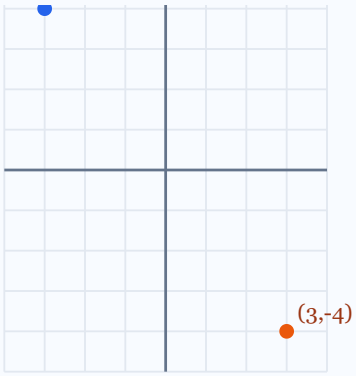
Check: both points are the same distance from the origin, $\sqrt{2^2 + 1^2} = \sqrt{5} = \sqrt{(-1)^2 + 2^2}$. ✓

Answer: $(-1, 2)$.

Example 2: 180° about the origin

Rotate $(3, -4)$ $180°$ about the origin.

Solution



180° is a straight-through turn — negate both coordinates.

A $180°$ rotation is a half-turn, equivalent to two $90°$ turns in a row, which negates both coordinates: $(x, y) \to (-x, -y)$.
$(3, -4) \to (-3, 4)$.
Answer: $(-3, 4)$.

Example 3: 270° counterclockwise about the origin

Rotate $(2, 5)$ $270°$ CCW about the origin.

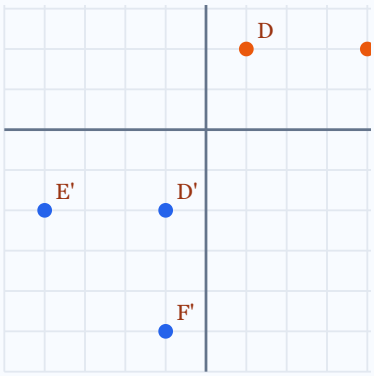
Solution

Rule: $270°$ CCW (the same as $90°$ clockwise) sends $(x, y) \to (y, -x)$.
$(2, 5) \to (5, -2)$.
Answer: $(5, -2)$.

Example 4: Rotate a whole triangle

$\triangle DEF$ has $D(1,2), E(4,2), F(1,5)$. Rotate $180°$ about the origin. Find all three image vertices.

Solution



Every vertex negates both coordinates under a 180° turn.

Apply $(x, y) \to (-x, -y)$ to each vertex: $D' = (-1, -2)$; $E' = (-4, -2)$; $F' = (-1, -5)$.
Answer: $D'(-1, -2), E'(-4, -2), F'(-1, -5)$.

Example 5: Why rotations preserve distance

Show algebraically that a $90°$ rotation about the origin preserves the distance between two points.

Solution

Let $P_1 = (x_1, y_1), P_2 = (x_2, y_2)$, rotated to $P_1' = (-y_1, x_1), P_2' = (-y_2, x_2)$.
The new coordinate differences are $-y_2 - (-y_1) = -(y_2 - y_1)$ and $x_2 - x_1$ — the same two differences as before, just swapped and one sign-flipped.
Since the distance formula squares both differences, the sign flip and the swap don't change the result: $(-(y_2 - y_1))^2 + (x_2 - x_1)^2 = (x_2 - x_1)^2 + (y_2 - y_1)^2$.
Answer: The distance is identical before and after, so rotations preserve distance.

Example 6: Orientation is preserved

Does a rotation reverse a figure's orientation, the way a reflection does? Explain.

Solution

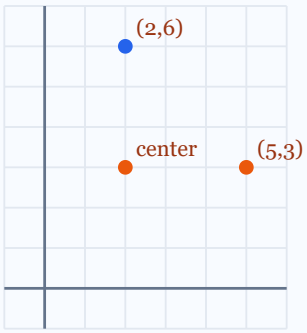
A rotation only turns a figure rigidly around a fixed point — it never flips the figure over, so the clockwise/counterclockwise order of its vertices stays the same.

Answer: No — rotations preserve orientation, along with translations; only reflections (and glide reflections) reverse it.

Example 7: Rotating about a point other than the origin

Rotate $(5, 3)$ $90°$ CCW about the center $(2, 3)$ (not the origin).

Solution



Shift the center to the origin, rotate, then shift back.

Step 1 — shift so the center is at the origin: subtract $(2, 3)$ from the point: $(5 - 2,\ 3 - 3) = (3, 0)$.

Step 2 — apply the $90°$ CCW rule $(x, y) \to (-y, x)$: $(3, 0) \to (0, 3)$.

Step 3 — shift back by adding $(2, 3)$: $(0 + 2,\ 3 + 3) = (2, 6)$.

Check: distance from center to $(5, 3)$ is 3; distance from center to $(2, 6)$ is $\sqrt{0^2 + 3^2} = 3$. ✓

Answer: $(2, 6)$.

Example 8: Identify the rotation from a point and its image

A point $(2, 3)$ maps to $(-3, 2)$ under a rotation about the origin. Which rotation is it?

Solution

Compare to the three rules: 90° CCW gives $(x, y) \to (-y, x)$, so $(2, 3) \to (-3, 2)$ — an exact match.

Answer: 90° counterclockwise (equivalently, 270° clockwise).

Example 9: Composing two 90° rotations

Rotate (x, y) by 90° CCW, then rotate that image by another 90° CCW. Show the result matches the single 180° rule.

Solution

First rotation: $(x, y) \to (-y, x)$.

Second rotation applied to $(-y, x)$: using the rule $(a, b) \to (-b, a)$ with $a = -y,\ b = x$, we get $(-x, -y)$.

Answer: $(-x, -y)$ — exactly the 180° rule, confirming that two 90° turns combine into one 180° turn (a preview of composing transformations in Section 3.4).

Example 10: From repeated rotation to rotational symmetry

A figure looks unchanged after being rotated 72° about its center. How many such rotations are needed to complete a full 360° turn, and what does this preview about Section 3.5?

Solution

Divide the full turn by the repeating angle: $360^\circ \div 72^\circ = 5$.

Preview: a figure that repeats every 72° has rotational symmetry of **order** 5 (like a regular pentagon) — Section 3.5 studies exactly this kind of repeat-angle counting.

Answer: 5 rotations of 72° complete the full turn.

Your Turn — Show Your Work!

For each problem, solve it and verify your solution. Show all steps clearly.

Problem 1: 90° CCW

Rotate $(3, 2)$ $90°$ CCW about origin.

Answer: **Verification:**

Problem 2: 180°

Rotate $(5, -1)$ $180°$ about origin.

Answer: **Verification:**

Problem 3: 270° CCW

Rotate $(1, 4)$ $270°$ CCW about origin.

Answer: **Verification:**

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Problem 4: 90° CW

Rotate $(2, 3)$ 90° CW about origin.

Answer: **Verification:**

Problem 5: Rule 90° CCW

$(x, y) \to ?$

Answer: **Verification:**

Problem 6: Rule 180°

$(x, y) \to ?$

Answer: **Verification:**

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Problem 7: Orientation

Rotation reverse orientation?

Answer: **Verification:**

Problem 8: Rigid?

Is rotation a rigid motion?

Answer: **Verification:**

Problem 9: 90° CCW

Rotate $(-2, 5)$ $90°$ CCW.

Answer: **Verification:**

Problem 10: 180°

Rotate $(-3, -6)$ $180°$.

Answer: **Verification:**

Problem 11: Equivalent

$270°$ CCW equals how many degrees CW?

Answer: **Verification:**

Problem 12: Center fixed

Does the center of rotation move?

Answer: **Verification:**

Problem 13: Congruent?

Figure congruent to its rotate?

Answer: **Verification:**

Problem 14: Challenge — rotate about another center

Rotate $(6, 4)$ $90°$ CCW about the center $(3, 4)$ (not the origin).

Answer: **Verification:**

Problem 15: Challenge — identify the rotation

A point $(4, -1)$ maps to $(1, 4)$ under a rotation about the origin. Which rotation is it?

Answer: **Verification:**

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✓ Answer Key

1. $(-2, 3)$
2. $(-5, 1)$
3. $(4, -1)$
4. $(3, -2)$
5. $(-y, x)$
6. $(-x, -y)$
7. No
8. Yes
9. $(-5, -2)$
10. $(3, 6)$
11. $90°$ CW
12. No
13. Yes
14. $(3, 7)$
15. $90°$ counterclockwise (rule $(x, y) \to (-y, x)$: $(4, -1) \to (1, 4)$)

Great work — keep practising!